GraphIV

Knowledge graph driven deep research tool

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Running order

1. Motivation
2. Method
3. Evaluation
4. Results
5. Discussion

SECTION 1 OF 5

1 Motivation

1 Motivation

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Research Question

A question I have at the start of projects:

What are people doing in (field) right now?

And what led them to doing it?

Is actually two questions in one:

- The **frontier** half wants what is recent and relevant.
- The **lineage** half wants the chain that leads to the frontier.

A list of papers answers neither!

Available Tools

AI search

One web search, retrieve whatever comes back, answer over it.

Shallow: As deep as whatever comes first

Deep research

Gemini, Claude, ChatGPT deep search that finds papers on its own.

No lineage: Finds papers. Cannot say which descends from which.

Consensus

The closest thing. Built to answer *what is the consensus in an area*.

Different question: Aggregates agreement, not development.

The gap: all three treat papers as independent sources. The citations/lineages that connects them gets neglected.

Goals

1. A tool that finds what people are doing now and *why* they are doing it given any research question
2. Local, self-hostable, open, and reliable. One consumer machine, local database, and every claim traceable.
3. A generalizable technique and findings.

Claim: Citations link related papers together. Knowing **why** *A* cites *B* explains the development in that field.
Graph retrieval is active: GraphRAG builds a graph over a corpus with an LLM [2]; G-Retriever answers over one textual graph [3]; PaSa is an agent that follows citations [4].
None fit here: the first cannot scale to 3.1M papers, the second targets a node's attributes, the third needs an API call per hop.

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2 Method

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The Corpus

Papers

3,073,288 arXiv papers, PDF and LaTeXML HTML on disk.

Intent, per edge

background | uses | extends | compares | critiques | motivates

Citations

176,934,085 occurrences parsed from HTML bodies.

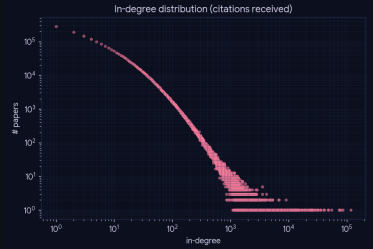
36,688,723 distinct in-corpus edges.

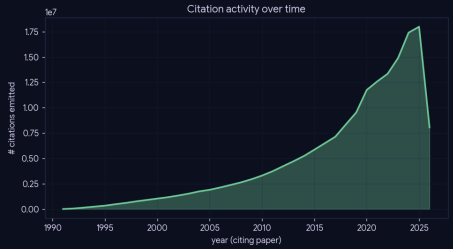
Embeddings

3.10M vectors, nomic-embed-text-v2-moe [5], pgvector HNSW.

Stock weights: nothing here is fine-tuned.

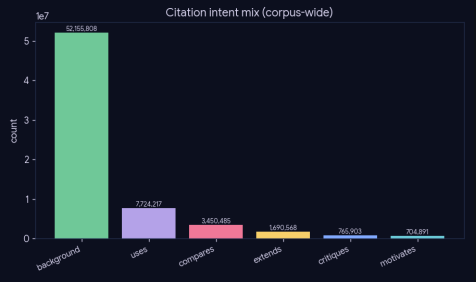
Corpus Shape





The tail is the problem: ResNet is cited 117,137 times inside the corpus, BERT 85,284, Attention 80,414. Rank on citation count and you return those for every question.

Edge Labels



Student, held out (*n*=13,521)

intent accuracy 0.844

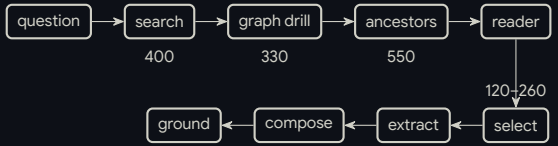
intent macro-F1 0.723

polarity accuracy 0.947

A Gemma teacher labels a sample, a DistilBERT student labels all 177M.

Weak classes: *uses* against *extends*, which are genuinely close in an author's phrasing.

Pipeline



Search only opens the shortlist. A vector search over abstracts gets 400 papers in. The drill then *cuts* to 330, and citations add the most: roughly a hundred papers a question arrive through the graph and are never found by search.

LeyLine

A **magnet** is a paper the reader rated highly. A walk starts where the most magnet mass is reachable and follows citations *forward*, cited to citing.

```
adj ← reverse(edges)
w ← {magnet: 1}
repeat MAX_LINEAGES:
   root ← argmax reachable_mass(adj, w)
   thread ← walk(root, k=1)
   collect magnets on thread; w[collected] *= GAMMA
return papers by collection frequency over SEEDS runs
```

k=1 makes a line, not a bush. **GAMMA** decay makes the next walk find a different story. It spends 8 of the landmark slots.

Joint Retrieval Method

The two signals, spent as one budget of 80 slots.

```
graph ← citations per year, as a percentile within the pool
reader ← relevance from −2..2, mapped onto 0..1
score ← 0.65 × graph  +  0.35 × reader
take the top 72 by score, give the last 8 to LeyLine
```

Citations per year, not raw count: a paper from last year with 200 citations beats a 2015 hub with 2000. Age is what stops the same five famous papers answering every question.

A paper the reader rejects is dropped from the ranking, but stays in the graph, since an off-topic paper can still be the link between two on-topic ones.

Generation Loop

1. **Grade**: the reader gives each candidate a role and a relevance in −2..2.
2. **Select**: landmarks, by survey or frontier mode.
3. **Extract**: a question-specific schema, filled per paper from full text.
4. **Compose**: a node table plus an intent-grouped adjacency list, so the writer sees relationships and not just papers.
5. **Judge**: a verdict decides whether to expand and retrieve again. It stops on its own.
6. **Ground**: every cited id is corpus-verified, typos corrected, fabrications stripped.

Steps 1 to 5 all run on one **Gemma-4-26B**, off the shelf, on the same 3090.

Step 6 is mechanical, not a prompt. A report cannot cite a paper that does not exist.

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3 Evaluation

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Survey-qrels

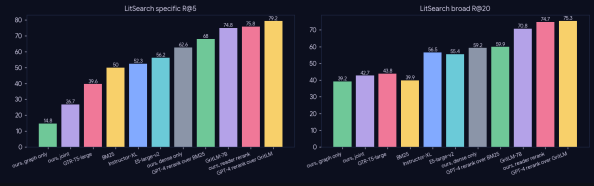
The construction

Take a survey paper. **Hide it.** Freeze the corpus to its publication month. Have an LLM write the question it answers.

The answer key is the survey's own reference list, filtered to works it discusses three or more times.

60 surveys, 2022–2024, 17 archives. 30 dev / 30 test, arms chosen on dev. 1,617 positive citations, 860 reachable at all. **Why not a relevance judgement**: nobody labels 3.1M papers. A survey's author already did the labelling, for their own field, with care no annotator will match.

LitSearch



597 queries [1]. Published rows are their table 3; ours measured through their index.

Reader rerank

75.8 specific R@5

74.7 broad R@20

Between GritLM-7B and GPT-4 reranking over it, on one 3090.

The caveat: LitSearch has 1.07 positive citations per query. It measures finding *one* paper, so it validates components, not the claim.

Support Check

The measurement

Every citing sentence is checked against *that paper's own text*, retrieving the passages nearest the claim. A sentence citing several papers passes if any one supports it.

The control

Break citations on purpose: swap the cited id for the **most similar paper the same report cites**. Same topic, so vocabulary alone will not catch it.

Then score the broken ones exactly like the real ones.

Two judge families, Gemma and Qwen. Gemma wrote the reports, so it is the one with a reason to flatter itself.

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4 Results

1 Motivation

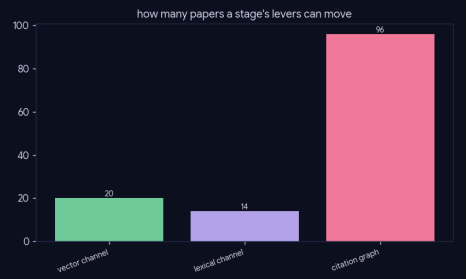
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Positive Citations, by Stage

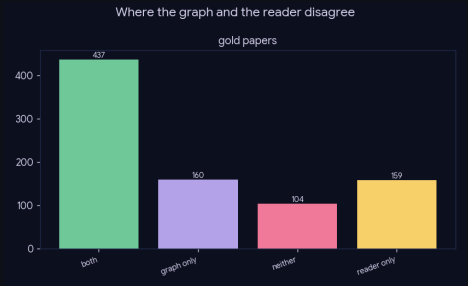


Eleven levers, grouped by stage. Spread:

positive citations moved, of 1,617	
vector channel	20
lexical channel	14
citation graph	**96**

The ceiling is set before ranking. 757 of 1,617 never reach the pool at all.

Two Ways to Spot a Paper



How the field cites it, or how a model reads it. Both beat chance by the same amount:

lift over base rate	
citations per year	2.89
reader relevance	2.88
text similarity	1.93

But they point at *different* papers: 160 only the graph finds, 159 only the reader. **So use both.**

With and Without the Graph

digest	coverage	supported	broken caught
graph	0.605	**0.707**	0.703
no graph	0.586	**0.754**	0.704

The graph does not help here. Same pool, same budget, edges 10,227 against 0. No-graph led in 8 of 10 matched pairs.

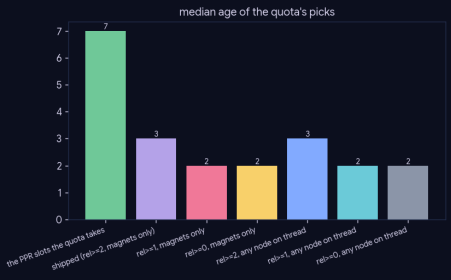
Untested hypothesis

Told *A cites B*, a writer produces *A builds on B*: true of the graph, unsupported by either paper.

How to Combine Them



Finding Newer Work



Of the papers LeyLine picks, how many did the survey actually cite?

picked papers that were cited by the survey	
LeyLine	**18.1%**
the slots it replaced	10.4%
picking at random	4.3%

Median age **3 years**, against 7.

Half the age, nearly twice as often right. That is the *what is happening now* half of the opening question.

Citation Approval



Ask a model: *does this paper actually say that?*

approved, of 818 sentences	
real citations	**7 in 10**
citations we broke	**under 2 in 10**

Two model families, agreeing on 90.6% of sentences.

The second number is what makes the first one mean anything. A checker that approves everything would score 10 in 10 on both.

What a Run Costs

Five topics, one instruction template, effort 4, one RTX 3090.

topic	wall	cited	verified	2026
neural audio codecs	6:03	45	45	11
test-time compute	2:46	56	56	5
state-space models	7:38	51	51	7
RL for theorem proving	5:14	60	60	12
QEC neural decoders	5:09	55	55	5
total		**267**	**267**	**40**

267 of 267 ids verified, 0 fabricated, and 97 papers from 2025 or later. Under eight minutes a question on one consumer GPU.

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5 Discussion

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Findings

1. **Retrieval is decided before ranking.** The pool stage moves 96 papers where the ranking rule moves 18.
2. **The graph and a reading model are equally strong and largely disjoint.** 2.89 against 2.88, with 160 papers only one sees and 159 only the other.
3. **Walking citations forward finds young work that lasts.** 18.1% against 10.4%, at a median age of 3 years against 7, using 8 slots of 80.
4. **Grounding is mechanical, faithfulness is not.** Every cited id exists by construction. Whether the paper supports the sentence is seven in ten.

Contributions

A dataset

Every arXiv paper and every citation between them, labelled with *why* one paper cites another. 3.1M papers, 177M edges.

An evaluation

Hide a survey, roll the corpus back to its month, ask its question, count how much of its reference list returns. 60 surveys, nobody labelled anything.

A retrieval method

LeyLine: walk citations forward from the papers a reader liked, and see where the threads land.

A composer

Turns the retrieved papers into a written review where every citation is checked against the corpus before it ships.

Open Problems

Forward impact

Every evaluation here is backward-looking. The question I want is whether a paper retrieved today *becomes* important.

Freeze at T, retrieve, label by citations accrued over two years, normalised against expected accrual, or it measures fame.

The uncited half

818 of 1,372 sentences carry a citation. Whether the rest invent anything is not measurable by prompting: two schemes failed a second judge, the stricter at $\kappa = 0.003$. **It needs human labels.**

Thank you

Demo: `graphiv.kuroma.dev`

References I

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G-retriever: Retrieval-augmented generation for textual graph understanding and question answering.
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References II

[4] Yichen He et al.
Pasa: An llm agent for comprehensive academic paper search.
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[5] Zach Nussbaum et al.
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